A determinant congruence for Heisenberg–Weyl blocks

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Abstract

Let R be a finite commutative local Frobenius ring of odd residue characteristic whose generating character ψ has order p — for example a finite field $\mathbb{F}_q$ or a truncated polynomial ring $\mathbb{F}_p[x]/(x^k)$ — and write $q = |R|$. Let H be the Heisenberg group of order q^3 over R and ψ_t a nontrivial character of its centre. An *inverse-closed section* c assigns to each nonzero $v \in R^2$ a central coordinate $c(v) \in R$ with $c(-v) = -c(v)$; the associated *Weyl block* $B_t(c)$ is the image, in the q-dimensional representation attached to ψ_t, of the sum of the corresponding group elements. We prove that the first-order variation of $\det B_t(c)$ about the zero section is divisible by $\lambda^{v_\lambda(q)+2}$, where $\lambda = 1 - \zeta_p$, and that the exponent splits canonically as

$$v_\lambda(q) \;+\; 2, \qquad \underbrace{v_\lambda(q)}_{\text{ramification of } q} \;+\; \underbrace{2}_{\text{inverse closure}}.$$

The proof rests on a closed formula $\operatorname{tr}\big(\operatorname{adj}(F)\,D\big) = \det(F)\,q\,S/(q^2-1)$ in which F is the block of the zero section and $S = \sum_v (\zeta_q^{-tc(v)} - 1)$; the block F is shown to satisfy $F^2 + 2F - (q^2-1)I = 0$ by a symplectic character sum. We then prove the congruence itself, in the form

$$\det B_t(c) \;\equiv\; \det F \pmod{\lambda^{\,v_\lambda(q)+4}},$$

valid for every such R and every section, modulo one explicitly identified lemma on the top exterior power. Here $v_\lambda(q) = \log_p(q)\,(p-1)$, so the exponent is $q+3$ when $R = \mathbb{F}_q$ with q prime, and 8 when $|R| = 9$. The constant 4 splits as $2+2$: two units of vanishing from inverse closure at first order, and two more from a cancellation between the first two orders whose mechanism is that a *symmetric* coefficient is paired against the *antisymmetric* symplectic form. Neither summand depends on the structure of R; only the ramification does. Several steps of the argument would fail on valuation counting alone, and each is rescued by inverse closure. The law is insensitive to whether R is a field: $\mathbb{F}_{p^k}$ and $\mathbb{F}_p[x]/(x^k)$ give the same exponent, while $\mathbb{Z}/p^2$ — whose generating character has order p^2 — does not satisfy the law at all. Exact certificates also attain the same exponent for two noncommutative Frobenius rings of order 81; those computations are evidence beyond the theorem's stated commutative-local scope, not a universal noncommutative theorem.

The congruence reduces to an identity about the power sums $\operatorname{tr}(D^m)$, and the main structural result of the paper concerns those. Expanding $\operatorname{tr}(D^m)$ over m-tuples of total displacement zero and grouping by cyclic orbit gives classes S_d indexed by the divisors of m. We prove a *sieve theorem*: if e denotes the order of the generating character and $t \mid m$ satisfies m/t odd and $e \mid (m/t)$, then

$$\sum_{d \mid t} d\,S_d \;=\; q\Big(\sum_{v \neq 0} d_v^{m/t}\Big)^t \;=\; 0.$$

Two apparently separate phenomena are its corollaries. When m is odd the index set of relations is downward closed, Möbius inversion runs, and each S_d with $e \mid (m/d)$ vanishes identically; when m is even it is not, and the theorem degenerates to a single relation. The sieve system is triangular along divisibility, so its rank is the number of relations and the number of unconstrained classes is $\tau(m) - \#\{u \mid m : u \text{ odd},\ e \mid u\}$, reduced by one more when $p \nmid m$ because then the period-one class is empty. Consequently $\mathrm{free}(1) = 0$, while $\mathrm{free}(m) = 1$ precisely when m is a positive power of p or a prime different from p. Finally $d\,S_d$ has the closed form $q \sum_{c \mid d} \mu(d/c)\bigl(\sum_v d_v^{m/c}\bigr)^c$, from which the sieve theorem follows in two lines. At $q = 3$, the complete section space realizes only six normalized cubic recurrences; their states modulo 9 prove $\min_c v_\lambda(\operatorname{tr} D_c^m) = 2(m + [m \text{ odd}])$ for every $m \geq 2$. This $q = 3$ theorem does not determine the minimum at $q \geq 5$.

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Status of the results

This paper records a programme in which one central conjecture was refuted and two statements built on it were corrected. Splitting the surviving mathematics from the corrections would make each half read more smoothly and would be dishonest: a paper cannot outsource its own errata. Instead, the status of every substantive claim is given here, once, before anything is proved.

claim	status
determinant congruence $\det B_t(c) \equiv \det F \pmod{\lambda^{v_\lambda(q)+4}}$	**stands.** Unconditional for $f \geq 2$; verified exhaustively at $q=3$ (Pass 473).
flat block quadratic $F^2 + 2F - (q^2-1)I = 0$	**stands.** Symplectic character sum.
first-order and sharp cancellation lemmas	**stand.**
sieve theorem $\sum_{d\mid t} d\, S_d = q\,\mathrm{Ps}(m/t)^t$	**stands.** Two independent proofs; the identity is also confirmed where its right side does *not* vanish.
odd-class vanishing, propagation, character-order form	**stand**, as corollaries of the sieve.
closed form $d\, S_d = q \sum_{c\mid d} \mu(d/c)\mathrm{Ps}(m/c)^c$	**stands.** Checked against honest enumeration.
transfer matrix $\mathrm{tr}(D^m) = q\,[T^m]_{0,0}$; $T \equiv 0 \bmod \lambda$	**stands**, and is now a corollary of translation covariance.
factorial law $v_\lambda(\mathrm{tr}\, D^m) = v_\lambda(q) + m + [m \text{ odd}] + v_\lambda(m!)$	**FALSE** at $q=3$ (Erratum 19); not attained among the sampled $q=5$ sections for $m = 10, 14, 15, 18, 19$, leaving the global minimum undecided.
replacement at $q=3$, $\min_c v_\lambda(\mathrm{tr}\, D_c^m) = 2(m + [m \text{ odd}])$	**stands for every** $m \geq 2$ (Theorem 21); $m=1$ is excluded because every section has trace zero.
sieve completeness ("nullity $= \lvert T\rvert$")	**CORRECTED CANDIDATE**: $\lvert T\rvert + [\,p \nmid m\,]$ is proved on the listed cells and two infinite families; the uniform composite-m case remains open (Erratum 26).
$\mathrm{free}(m) = 1 \iff m = p^j$	**CORRECTED**: also every prime $\ell \neq p$ (Correction 31).

All three failures share one cause, and it is not carelessness: each was confirmed only on cases that were cheap to compute, and cheapness was governed by the same condition that made the claim true there. That failure mode is analysed separately in the companion note `papers/agreement_locus.tex`; readers interested in the methodology rather than the arithmetic may prefer to start there.

1 Introduction

Let R be a finite commutative local Frobenius ring of odd residue characteristic p whose generating character ψ has order p, and put $q = \lvert R\rvert$. Two families of examples run through the paper: the finite fields $R = \mathbb{F}_q$, and the truncated polynomial rings $R = \mathbb{F}_p[x]/(x^k)$, whose socle is (x^{k-1}) and whose generating character is $\psi(c) = \zeta_p^{c_{k-1}}$. The reader who prefers may take $R = \mathbb{F}_q$ throughout; nothing in the argument uses invertibility of nonzero elements, and Section 5 shows the distinction is invisible to the result. The hypothesis is not ad hoc: by Wood's theorem [3] the finite Frobenius rings are precisely the finite rings for which the MacWilliams extension theorem holds, and a generating character is the standard tool in that theory, so the setting here is the usual one of ring-linear coding theory. What our argument additionally requires — and what fails over $\mathbb{Z}/p^2$ — is only that the generating character have order p. The theorem statements below retain this commutative-local scope. The two noncommutative examples in Section 5 are exact computational extensions and are labelled as such. Set

$$H = \{(a,b,c) : a,b,c \in R\}, \qquad (a,b,c)(a',b',c') = (a+a',\, b+b',\, c+c'-ab'+a'b),$$

a group of order q^3 with centre $Z = \{(0,0,c)\}$; for $R = \mathbb{F}_q$ it is the extraspecial group of exponent q. Each nontrivial character $\psi_t(0,0,c) = \zeta^{tc}$ ($\zeta = e^{2\pi i/q}$, $t \not\equiv 0$) determines a q-dimensional irreducible representation ρ_t of H, realized on $\mathbb{C}^{\mathbb{F}_q}$ by

$$\rho_t(a,b,c)\, e_x \;=\; \zeta^{\,t(c+2xb+ab)}\, e_{x+a}. \tag{1}$$

Call a function $c : \mathbb{F}_q^2 \setminus \{0\} \to \mathbb{F}_q$ an *inverse-closed section* if $c(-v) = -c(v)$; equivalently, the set $S(c) = \{(v, c(v)) : v \neq 0\} \subset H$ is closed under inversion, since $(a,b,z)^{-1} = (-a,-b,-z)$. Such sections index the ways of lifting the punctured plane $\mathbb{F}_q^2 \setminus \{0\}$ into H compatibly with inversion, and the object of interest is the *Weyl block*

$$B_t(c) \;=\; \sum_{v \in \mathbb{F}_q^2 \setminus \{0\}} \rho_t\big(v, c(v)\big) \;\in\; M_q\big(\mathbb{Z}[\zeta]\big),$$

a Hermitian matrix over the cyclotomic ring $\mathbb{Z}[\zeta]$. Write $F = B_t(0)$ for the block of the zero (*flat*) section.

Sections arise naturally: $S(c)$ together with $Z \setminus \{1\}$ is the Cayley set of a strongly regular graph on H with parameters $\big(q^3, (q-1)(q+2), q-2, q+2\big)$ exactly when c is flat up to the obvious symmetries, and the family of these partial difference sets in H is classical [1, 2]. The determinants $\det B_t(c)$, as c ranges over all sections, are therefore a natural arithmetic invariant of that family. Our results describe how little they vary.

Throughout, $\lambda = 1 - \zeta$ is the unique prime of $\mathbb{Z}[\zeta]$ above q, so that $(q) = (\lambda)^{q-1}$ and

$$v_\lambda(q) = q - 1, \tag{2}$$

and v_λ denotes the λ-adic valuation.

Theorem 1 (first-order law)**.** *Let q be an odd prime, $t \not\equiv 0$, and c an inverse-closed section. Put $D = B_t(c) - F$ and $T_1 = \operatorname{tr}\big(\operatorname{adj}(F)\, D\big)$, the first-order term of* $\det(F + D)$*. Then*

$$T_1 \;=\; \frac{\det(F)\; q\; S}{q^2 - 1}, \qquad S = \sum_{v \in \mathbb{F}_q^2 \setminus \{0\}} \big(\zeta^{-tc(v)} - 1\big),$$

and consequently

$$v_\lambda(T_1) \;=\; (q-1) + v_\lambda(S) \;\geq\; (q-1) + 2 \;=\; q + 1.$$

The two summands are structurally distinct: $q - 1 = v_\lambda(q)$ is the ramification index (2), while the 2 comes from inverse closure through the factorization $w + w^{-1} - 2 = -(1-w)(1-w^{-1})$ applied to $w = \zeta^{tc(v)}$. Only the first depends on q being prime; writing the bound as $v_\lambda(q) + 2$ makes the statement correct for every prime power (Section 5).

Section 4 proves the congruence at exponent $v_\lambda(q) + 2$ (Theorem 12) and then lifts it to the sharp $v_\lambda(q) + 4$ (Theorems 16 and 17), modulo a single explicitly identified bound on the top exterior power. Section 5 explains why no separate prime-power statement is needed.

Verification and reproducibility

All computations reported here are exact: cyclotomic integers are represented in the basis $1, \zeta, \dots, \zeta^{q-2}$ and valuations are read off from $v_\lambda(x) = v_q\big(N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(x)\big)$. Witness programs and their certificates are archived in the repository [5]; the pairing identity of Lemma 9 is machine-checked in Lean 4 against Mathlib [4].

1.1 How the results depend on one another

The sections below follow the order in which the results were found, which is not the order in which they depend on each other. A reader who wants the logical structure should take it as follows.

At the root is a single lemma about one power sum. Writing $\mathrm{Ps}(k) = \sum_{v \neq 0} d_v^k$ and letting e be the order of the generating character,

$$\mathrm{Ps}(k) = 0 \text{ for every inverse-closed section } \iff k \text{ is odd and } e \mid k. \tag{$*$}$$

Inverse closure pairs v with $-v$, so $\mathrm{Ps}(k)$ is a sum of terms $2\,\mathrm{Re}(u-1)^k$, and $(u-1)^k$ is purely imaginary exactly under those conditions; the one-pair section of Proposition 23 supplies the converse.

Everything else descends from $(*)$. The closed form (Proposition 27) writes each cyclic class $d\,S_d$ in terms of Ps alone; summing it over $d \mid t$ and exchanging the order gives the sieve theorem (Theorem 24), whose right-hand side is $q\,\mathrm{Ps}(m/t)^t$ and therefore vanishes by $(*)$. The sieve in turn yields the odd-class vanishing theorem (Theorem 22) when its index set is downward closed, and a single propagation relation when it is not; its system is triangular along divisibility, which gives the rank count (Proposition 25) and hence $\mathrm{free}(m) = 1 \iff m$ is a positive power of p or a prime different from p, with $\mathrm{free}(1) = 0$ (Proposition 30). The determinant congruence of §4, which motivated all of this, sits at the other end: it follows from the factorial law, which is an assertion about the very power sums $(*)$ constrains.

One boundary is worth stating at the outset. The closed form requires $e \mid (m/d)$, and the top class $d = m$ never satisfies it; the reason is identified in §4.4, and it is a genuine feature of the object rather than a defect of the argument.

2 The flat block

Lemma 2. *For every noncentral $g \in H$ and every $t \not\equiv 0$, $\operatorname{tr}\rho_t(g) = 0$. Consequently $\operatorname{tr} F = 0$ and $\operatorname{tr} B_t(c) = 0$ for every section c.*

Proof. Write $g = (a, b, c)$. If $a \neq 0$ then by (1) the matrix $\rho_t(g)$ moves every basis vector e_x to $e_{x+a} \neq e_x$, so its diagonal vanishes. If $a = 0$ and $b \neq 0$ then $\rho_t(0, b, c)$ is diagonal with entries $\zeta^{t(c+2xb)}$, and $\operatorname{tr}\rho_t(0, b, c) = \zeta^{tc}\sum_{x \in \mathbb{F}_q}\zeta^{2tbx} = 0$ because $2tb \neq 0$ in $\mathbb{F}_q$ (q is odd). Every element of $S(c)$ is noncentral, whence the last claim. $\square$

Let $\omega(v, w) = ab' - a'b$ denote the symplectic form on $\mathbb{F}_q^2$, where $v = (a, b)$ and $w = (a', b')$, so that $(v, 0)(w, 0) = (v + w,\ -\omega(v, w))$.

Lemma 3 (flat block)**.** *$F^2+2F-(q^2-1)I=0$. Hence F has spectrum $\{(q-1)^{(q+1)/2}, (-(q+1))^{(q-1)/2}\}$,*

$$\det F=(q-1)^{(q+1)/2}\big(-(q+1)\big)^{(q-1)/2}, \qquad F^{-1}=\frac{F+2I}{q^2-1},$$

and $\det F$ is a unit at λ.

Proof. Expanding and using $\rho_t(u,z)=\zeta^{tz}\rho_t(u,0)$,

$$F^2=\sum_{v,w\neq 0}\rho_t\big((v,0)(w,0)\big)=\sum_{v,w\neq 0}\zeta^{-t\,\omega(v,w)}\,\rho_t(v+w,0).$$

Split according to $u=v+w$. The terms with $u=0$ have $w=-v$ and $\omega(v,-v)=0$, contributing $(q^2-1)I$. For fixed $u\neq 0$ the constraint is $v\notin\{0,u\}$, and $\omega(v,u-v)=\omega(v,u)$, so those terms contribute

$$\Big(\sum_{v\neq 0,u}\zeta^{-t\,\omega(v,u)}\Big)\rho_t(u,0).$$

Since $u\neq 0$, the map $v\mapsto\omega(v,u)$ is a nonzero linear functional on $\mathbb{F}_q^2$, so $\sum_{v\in\mathbb{F}_q^2}\zeta^{-t\omega(v,u)}=0$; removing the two excluded values $v=0$ and $v=u$, both of which give $\omega=0$, leaves -2. Summing over $u\neq 0$ gives $-2F$, so $F^2=(q^2-1)I-2F$.

Thus the minimal polynomial of F divides $x^2+2x-(q^2-1)=(x-(q-1))(x+(q+1))$, so the eigenvalues lie in $\{q-1,-(q+1)\}$. If they occur with multiplicities α,β then $\alpha+\beta=q$ and, by Lemma 2, $\alpha(q-1)-\beta(q+1)=0$; solving gives $\alpha=(q+1)/2$, $\beta=(q-1)/2$. The remaining assertions follow, the last because $q\pm 1$ are prime to q. □

Remark 4. Lemma 3 also reproves the two "trace laws" $\operatorname{tr}F=0$ and $\operatorname{tr}F^2=q(q^2-1)$; the latter is $(q-1)^2\frac{q+1}{2}+(q+1)^2\frac{q-1}{2}=q(q^2-1)$. These hold for every section, not only the flat one, and are used below.

Remark 5 (the flat block as a quadratic order)**.** Completing the square in Lemma 3 is instructive. Put $S=F+q+1$; then $F^2+2F-(q^2-1)=0$ becomes

$$S^2-2qS=0, \qquad S\in\{0,\,2q\},$$

so $\mathbb{Z}[F]$ is the quadratic order $O_q=\mathbb{Z}[S]/\big(S(S-2q)\big)$, a node with two branches whose root gap is the eigenvalue gap $2q$. Its conductor in the normalisation $\mathbb{Z}\times\mathbb{Z}$ is $2q$, and over $\mathbb{Z}_p$ the two branch lattices M_0, M_{2q} (with S acting as 0 and $2q$) have the periodic resolution $\cdots\to O_q\xrightarrow{S-2q}O_q\xrightarrow{S}O_q\to M_0$, giving the complete Ext quiver

$$\big(\mathrm{Ext}^1_{\mathrm{self}},\mathrm{Ext}^1_{\mathrm{cross}},\mathrm{Ext}^2_{\mathrm{self}},\mathrm{Ext}^2_{\mathrm{cross}}\big)=\big(0,\ \mathbb{Z}/p^{v_p(2q)},\ \mathbb{Z}/p^{v_p(2q)},\ 0\big),$$

the kernel entries vanishing over the torsion-free $\mathbb{Z}_p$; the quadratic Kuranishi obstruction for $M_0\oplus M_{2q}$ is $(x,y)\mapsto(xy,xy)$, with unobstructed cone $xy=0$. At $q=2$, where $v_2(2q)=v_2(4)=2$, this is exactly the $S8$ two-character commutant $\mathbb{Z}_2[S]/(S^2-4S)$ with $\mathrm{Ext}=\mathbb{Z}/4$ and cone $xy=0$ studied in the companion 2-adic deformation work; the flat block is its odd-q family, and the odd-prime members are predicted to have $\mathrm{Ext}=\mathbb{Z}/q$. Whether the $S8$ characteristic lattices *are* these branch lattices, rather than only sharing their homological invariants, is left open.

Remark 6 (the flat block's eigenlattice gluing)**.** The Ext quiver of Remark 5 is an invariant of the *abstract* order O_q; the flat block realises that order on the concrete lattice $\mathbb{Z}^n$, $n = q(q-1)$, obtained by writing the $q \times q$ operator over $\mathbb{Z}[\zeta_q]$ as an integer operator, and one may ask how its two *saturated* eigenlattices $L_{q-1} = \ker(F - (q-1)I)$ and $L_{-(q+1)} = \ker(F + (q+1)I)$ glue inside it. Since the eigenvalue gap is $2q$, $P = (F + (q+1)I)/2q$ is the spectral projector onto L_{q-1}, and a vector is integral in $L_{q-1} \oplus L_{-(q+1)}$ exactly when Pv is, i.e. when $(F+(q+1)I)v \equiv 0 \pmod{2q}$; hence by Proposition 8

$$\mathbb{Z}^n/(L_{q-1} \oplus L_{-(q+1)}) \;=\; \operatorname{im}\big((F + (q+1)I) \bmod 2q\big) \;\cong\; (\mathbb{Z}/2)^{(q-1)^2/2},$$

pure 2-torsion, verified exactly at $q = 3, 5, 7$ as $(\mathbb{Z}/2)^2, (\mathbb{Z}/2)^8, (\mathbb{Z}/2)^{18}$. The exponent $(q-1)^2/2$ is the $\mathbb{Z}$-rank of the smaller eigenlattice $L_{-(q+1)}$. The gluing is 2-torsion because modulo q one has $F + (q+1)I \equiv F + I$ with $(F + I)^2 \equiv F^2 + 2F + 1 \equiv 0$, so every Smith invariant of $F + (q+1)I$ is divisible by q and $2q/\gcd(d_i, 2q) \in \{1, 2\}$; the conductor's odd part contributes nothing to a lattice on which q is already absorbed. There is no q-torsion in the gluing, and in particular no relation to the antipodal-pair count. This realisation is field-specific: over the ring $\mathbb{Z}/9$ the flat block has $q = 9$ but fails $F^2 + 2F - (q^2 - 1)I = 0$ in every entry and has neither $q - 1 = 8$ nor $-(q+1) = -10$ as an eigenvalue, so no two-branch order describes it for $n > 1$.

Erratum 7. An earlier version of Remark 6 reported the gluing as $(\mathbb{Z}/2q)^{q-1} \oplus (\mathbb{Z}/q)^{(q^2-1)/2-(q-1)}$ with q-primary rank $(q^2-1)/2$, and inferred a "deformation–Burnside bridge" identifying that rank with the antipodal-pair count. Both are withdrawn: that computation glued the eigenlattice *images* rather than the saturated eigenlattices and used a faulty integer Smith routine. The correct gluing is the pure 2-torsion group above, whose rank $(q-1)^2/2$ never equals $(q^2-1)/2$; there is no such bridge.

Proposition 8 (two-branch gluing)**.** *Let S be an integral operator on $\mathbb{Z}^n$ with $S(S - cI) = 0$, $c > 0$. In a unimodular basis adapting the saturated c-eigenlattice, write $S = \left[\begin{smallmatrix} cI_a & Y \\ 0 & 0 \end{smallmatrix}\right]$ with Y an integer $a \times b$ block, a the rank of the c-branch and b that of the 0-branch. Then the gluing of the two saturated eigenlattices is*

$$\mathbb{Z}^n/(L_c + L_0) \;=\; \operatorname{im}(Y \bmod c) \;\cong\; \bigoplus_i \mathbb{Z}/\big(c/\gcd(d_i, c)\big),$$

where $d_1, \ldots, d_{\min(a,b)}$ are the Smith invariants of Y.

Proof. Modding out $L_c = \mathbb{Z}^a \oplus 0$ collapses the first a coordinates, so $\mathbb{Z}^n/(L_c + L_0) = \mathbb{Z}^b/\pi_y(L_0)$, and $\pi_y(L_0) = \{y : Yy \equiv 0 \pmod c\} = \ker(Y \bmod c)$ because L_0 is saturated; hence the quotient is $\operatorname{im}(Y \bmod c)$. Writing $Y = UDV$ with U, V unimodular (Smith form), U, V stay invertible modulo c, so $\operatorname{im}(Y \bmod c) \cong \operatorname{im}(D \bmod c) = \bigoplus_i d_i(\mathbb{Z}/c\mathbb{Z}) = \bigoplus_i \mathbb{Z}/(c/\gcd(d_i, c))$. □

Every cyclic factor divides the conductor c; the top factor is c exactly when some d_i is coprime to c, and the number of full-conductor $\mathbb{Z}/c$ summands is $\#\{i : \gcd(d_i, c) = 1\}$. Thus the conductor is intrinsic to the abstract order while the whole gluing type is a property of the realisation. For the cyclotomic flat block at gap $c = 2q$ (Remark 6) no d_i is coprime to c: the gluing is the pure 2-torsion $(\mathbb{Z}/2)^{(q-1)^2/2}$, with zero full-conductor summands. On the $W(3, 3)$ signed-turn lattices, by contrast, the cycle lattice at $c = 4$ has off-diagonal Smith type $(1^{66}, 12)$, giving 66 full-conductor summands and gluing $(\mathbb{Z}/4)^{66}$, while the cut lattice at $c = 6$ has 10.

So a single gap-c order supports gluings as different as $(\mathbb{Z}/2)^2$, $(\mathbb{Z}/4)^{66}$ and $(\mathbb{Z}/2)^5 \oplus (\mathbb{Z}/6)^{10}$, according to the Smith type of Y alone.

For an operator with $k > 2$ distinct integer eigenvalues $c_1, \dots, c_k$ the same argument, applied to the spectral projectors $P_i = N_i/D_i$ with $N_i = \prod_{j \neq i}(S - c_j I)$ and $D_i = \prod_{j \neq i}(c_i - c_j)$, gives

$$\mathbb{Z}^n / \bigoplus_i L_i \;=\; \mathbb{Z}^n / \bigcap_i \ker\bigl(N_i \bmod D_i\bigr),$$

now supported on the several products D_i rather than a single conductor; a closed form in the D_i is open.

The substrate supplies such an operator. The signed-turn operator $K = R^T(2T - B)R$ on the 240 integral edge chains of $W(3,3)$ satisfies $(K + 6I)(K - 2I)(K - 4I)(K - 10I) = 0$ exactly over $\mathbb{Z}$, with eigenspace dimensions $81, 120, 24, 15$. Its four saturated eigenlattices glue to

$$\mathbb{Z}^{240} / \bigoplus_i L_i \;\cong\; (\mathbb{Z}/32)^{14} \oplus (\mathbb{Z}/8) \oplus (\mathbb{Z}/4)^{66} \oplus (\mathbb{Z}/2)^{23} \oplus (\mathbb{Z}/3)^{10} \oplus (\mathbb{Z}/5)^{23}.$$

Two summands are visible on the two-branch restrictions computed independently elsewhere: the cycle lattice, where $S = (K + 6I)/2$ satisfies $S(S - 4) = 0$ and the off-diagonal block has Smith type $(1^{66}, 12)$, contributes the $(\mathbb{Z}/4)^{66}$; and the cut lattice, where $S = K - 4I$ satisfies $S(S - 6) = 0$, contributes the $(\mathbb{Z}/3)^{10}$, whose exponent is $\Phi_4(3) = 10$. So the two-branch restrictions are sub-objects of a single four-branch gluing.

3 The first-order law

Lemma 9 (pairing)**.** *Let R be a commutative ring and $w \in R$ a unit with inverse w^{-1}. Then*

$$w + w^{-1} - 2 = -(1 - w)(1 - w^{-1}).$$

In particular, if $\mu \mid (1 - w)$ and $\mu \mid (1 - w^{-1})$ then $\mu^2 \mid \bigl(w + w^{-1} - 2\bigr)$.

Proof. $(1 - w)(1 - w^{-1}) = 1 - w - w^{-1} + ww^{-1} = 2 - w - w^{-1}$. $\square$

Proof of Theorem 1. Since $\det F$ is a λ-unit, F is invertible and $T_1 = \operatorname{tr}(\operatorname{adj}(F)D) = \det(F)\operatorname{tr}(F^{-1}D)$. By Lemma 3, $F^{-1} = (F + 2I)/(q^2 - 1)$, and $\operatorname{tr} D = \operatorname{tr} B_t(c) - \operatorname{tr} F = 0$ by Lemma 2, so

$$T_1 = \frac{\det(F)}{q^2 - 1}\Bigl[\operatorname{tr}(FD) + 2\operatorname{tr} D\Bigr] = \frac{\det(F)\ \operatorname{tr}(FD)}{q^2 - 1}.$$

Now $\operatorname{tr}(FD) = \operatorname{tr}\bigl(FB_t(c)\bigr) - \operatorname{tr}(F^2) = \operatorname{tr}\bigl(FB_t(c)\bigr) - q(q^2 - 1)$. For the first term,

$$\operatorname{tr}\bigl(FB_t(c)\bigr) = \sum_{v,w \neq 0} \operatorname{tr}\rho_t\bigl((v,0)(w,c(w))\bigr), \qquad (v,0)\bigl(w,c(w)\bigr) = \bigl(v + w,\ c(w) - \omega(v,w)\bigr).$$

By Lemma 2 only central products contribute, i.e. only $w = -v$; then $\omega(v, -v) = 0$ and the product is $\bigl(0, 0, c(-v)\bigr) = (0, 0, -c(v))$, whose trace is $q\,\zeta^{-tc(v)}$. Hence $\operatorname{tr}(FB_t(c)) = q\sum_{v \neq 0}\zeta^{-tc(v)}$, and since there are $q^2 - 1$ nonzero v,

$$\operatorname{tr}(FD) = q\sum_{v \neq 0}\bigl(\zeta^{-tc(v)} - 1\bigr) = q\,S,$$

which is the displayed formula.

For the valuation, $v_\lambda(\det F) = v_\lambda(q^2-1) = 0$ and $v_\lambda(q) = q-1$ by (2), so $v_\lambda(T_1) = (q-1)+v_\lambda(S)$. Group the sum defining S into the $(q^2-1)/2$ pairs $\{v,-v\}$. Using $c(-v) = -c(v)$ and Lemma 9 with $w = \zeta^{tc(v)}$,

$$\big(\zeta^{-tc(v)} - 1\big) + \big(\zeta^{tc(v)} - 1\big) = -\big(1-\zeta^{tc(v)}\big)\big(1-\zeta^{-tc(v)}\big).$$

Each factor is 0 if $c(v) = 0$ and has valuation 1 otherwise, so every pair contributes valuation 2 or ∞; therefore $v_\lambda(S) \geq 2$ and $v_\lambda(T_1) \geq q+1$. □

4 The congruence

Write $G = F^{-1}D$, $p_m = \operatorname{tr}(G^m)$, and $e_k = e_k(G)$, so that $\det B_t(c) = \det(F)\big(1+\sum_{k\geq 1} e_k\big)$. Note G is λ-integral, since $F^{-1} = (F+2I)/(q^2-1)$ and q^2-1 is prime to λ; hence so is every e_k.

Lemma 10 (power sums)**.** *$v_\lambda(p_m) \geq q+1$ for every $m \geq 1$.*

Proof. Put $H = (F+2I)D$, so $p_m = \operatorname{tr}(H^m)/(q^2-1)^m$ and $v_\lambda(p_m) = v_\lambda(\operatorname{tr} H^m)$. Both F and D are $\mathbb{Z}[\zeta]$-linear combinations of the matrices $\rho_t(g)$, and $D = \sum_{v\neq 0} d_v \rho_t(v,0)$ with $d_v = \zeta^{tc(v)}-1$, each of valuation ≥ 1. Expanding $\operatorname{tr}(H^m)$ therefore produces monomials $d_{v_1}\cdots d_{v_m}\cdot \operatorname{tr}\rho_t(g)$ for single group elements g, with exactly m factors d. By Lemma 2 such a trace vanishes unless g is central, in which case it equals q times a root of unity. Every surviving monomial thus carries one factor q and m factors d, so $v_\lambda(\operatorname{tr} H^m) \geq (q-1)+m$, which is $\geq q+1$ for $m \geq 2$.

For $m = 1$ this count gives only q. But by the computation in the proof of Theorem 1, $\operatorname{tr} H = \operatorname{tr}(FD) + 2\operatorname{tr} D = qS$, and inverse closure gives $v_\lambda(S) \geq 2$; hence $v_\lambda(p_1) = (q-1) + v_\lambda(S) \geq q+1$ as well. □

Lemma 11 (top exterior power)**.** *$v_\lambda(e_q) \geq q+1$.*

Proof. $e_q = \det G = \det D/\det F$ and $v_\lambda(\det F) = 0$, so it suffices that $v_\lambda(\det D) \geq q+1$. Every entry of D is a $\mathbb{Z}[\zeta]$-combination of the d_v, so $D = \lambda D'$ with D' λ-integral and $\det D = \lambda^q \det D'$; we must show $\det D' \equiv 0 \pmod{\lambda}$.

Reduce modulo λ, identifying $\mathbb{Z}[\zeta]/\lambda = \mathbb{F}_q$ via $\zeta \mapsto 1$. Then $(\zeta^k-1)/(1-\zeta) \equiv -k$ and $\rho_t(v,0) \equiv P_{a_v}$, where P_a is the permutation matrix of $x \mapsto x+a$; grouping the q^2-1 vectors by their first coordinate,

$$D' \;\equiv\; \sum_{a\in\mathbb{F}_q} \kappa_a P_a \pmod{\lambda}, \qquad \kappa_a = -t\sum_b c(a,b).$$

The P_a span the group algebra $\mathbb{F}_q[\mathbb{Z}/q] = \mathbb{F}_q[y]/(y^q-1) = \mathbb{F}_q[y]/\big((y-1)^q\big)$, in which the determinant of multiplication by $f(y)$ is $f(1)^q$. Hence $\det D' \equiv \big(\sum_a \kappa_a\big)^q = \big(-t\sum_{v\neq 0} c(v)\big)^q$, and inverse closure makes $\sum_{v\neq 0} c(v) = 0$ because the terms cancel in pairs $\{v,-v\}$. So $\det D' \equiv 0$. □

Theorem 12 (the congruence)**.** *For every odd prime power q, every $t \not\equiv 0$, and every inverse-closed section c,*

$$\det B_t(c) \;\equiv\; \det F \pmod{\lambda^{v_\lambda(q)+2}}.$$

Proof. By Newton's identities, $k\,e_k = \sum_{i=1}^{k}(-1)^{i-1}e_{k-i}p_i$. For $1 \le k \le q-1$ the integer k is a unit at λ, so induction on k together with Lemma 10 and the λ-integrality of the e_j gives $v_\lambda(e_k) \ge q+1$. For $k = q$ the same recursion would lose $v_\lambda(q) = q-1$, and Lemma 11 supplies that case directly. Hence $v_\lambda(e_k) \ge q+1$ for all $k \ge 1$ and $\det B_t(c) - \det F = \det(F)\sum_{k\ge1} e_k$ has valuation $\ge q+1$. □

Remark 13. Both places where valuation counting alone is insufficient — the first power sum in Lemma 10 and the top exterior power in Lemma 11 — are rescued by inverse closure, in the first case through $v_\lambda(S) \ge 2$ and in the second through $\sum_v c(v) = 0$. Without that hypothesis the argument gives only λ^q.

4.1 The sharp exponent

The exponent can be improved by exactly two, and the improvement is a single cancellation which we now prove.

Lemma 14 (parity)**.** *For odd m, $v_\lambda(p_m) \ge v_\lambda(q) + m + 1$. In particular $v_\lambda(p_3) \ge v_\lambda(q) + 4$.*

Proof. By the proof of Lemma 10 the coefficient of the leading power of λ in $\mathrm{tr}(H^m)$ is, up to a unit, a sum over m-tuples $(v_1, \dots, v_m)$ of total displacement zero of $m_{v_1}\cdots m_{v_m}$, where $d_v \equiv -m_v\lambda$ and $m_v = \psi$-exponent of $c(v)$. The involution $(v_1, \dots, v_m) \mapsto (-v_1, \dots, -v_m)$ preserves the constraint, and $c(-v) = -c(v)$ sends each m_{v_i} to $-m_{v_i}$, multiplying the summand by $(-1)^m = -1$. So the sum equals its own negative and vanishes in the residue field, gaining one further power of λ. □

Lemma 15 (the two closed forms)**.** *Put $Q = \sum_{v,w\ne0} d_v d_w\, \psi_t\bigl(-\omega(v,w)\bigr)$ and $R = \mathrm{tr}(FDFD)/q$. Then, exactly,*

$$\mathrm{tr}\,H = qS, \quad \mathrm{tr}(D^2) = -2qS, \quad \mathrm{tr}(FD^2) = q\,(Q + 2S), \quad \mathrm{tr}(H^2) = qR + 4qQ,$$

and, modulo λ^4, $Q \equiv 0$ and $R \equiv -2S$.

Proof. The exact identities follow as in Theorem 1, keeping track of which products of ρ's are central. For Q: expanding $\psi_t(-\omega) \equiv 1 + (\text{linear in } \lambda)$ and using $d_v d_w \equiv m_v m_w \lambda^2$, the constant term is $\sum_{v,w} d_v d_w = S^2$, of valuation ≥ 4 since $v_\lambda(S) \ge 2$; the surviving term pairs the symmetric coefficient $d_v d_w$ against the antisymmetric $\omega(v,w)$ and therefore cancels under $(v,w) \mapsto (w,v)$. For R: reparametrizing so that the two free vectors are summed for fixed (v,w), and writing $s = -(v+w)$, the phase is $2\omega(x,v) + \omega(x,s) + \omega(v,s)$; summing over the q^2-2 admissible x and using $\sum_{x\in\mathbb{F}_q^2}\omega(x,v) = 0$ gives $2\omega(v,s) + 0 + (q^2-2)\omega(v,s) = q^2\omega(v,s)$, which vanishes in the residue field. What survives is the count of admissible pairs, giving $R \equiv -2S$. □

Theorem 16 (the cancellation)**.** $2p_1 \equiv p_2 \pmod{\lambda^{v_\lambda(q)+4}}$*; equivalently* $e_1 + e_2 \equiv 0 \pmod{\lambda^{v_\lambda(q)+4}}$.

Proof. By Lemma 15,

$$2(q^2-1)\,\mathrm{tr}\,H - \mathrm{tr}(H^2) = q\bigl[\,2(q^2-1)S - R - 4Q\,\bigr] \equiv q\bigl[\,2(q^2-1)S + 2S\,\bigr] = q\cdot 2q^2 S \pmod{q\lambda^4},$$

and $v_\lambda(2q^2S) = 2v_\lambda(q) + v_\lambda(S) \ge 4$. Dividing by $(q^2-1)^2$, a λ-unit, gives $v_\lambda(2p_1 - p_2) \ge v_\lambda(q) + 4$. Finally $v_\lambda(p_1^2) \ge 2(v_\lambda(q)+2) \ge v_\lambda(q) + 4$, so $e_2 = (p_1^2 - p_2)/2$ yields $e_1 + e_2 \equiv p_1 - p_2/2 \equiv 0$. □

Theorem 17 (the sharp law)**.** *Suppose $v_\lambda(e_q) \geq v_\lambda(q)+4$. Then for every odd prime power $q = p^f$,*

$$\det B_t(c) \ \equiv\ \det F \pmod{\lambda^{v_\lambda(q)+4}}.$$

The hypothesis is automatic whenever $f \geq 2$, so for every odd prime power that is not prime the conclusion is unconditional.

Proof that the hypothesis is automatic for $f \geq 2$. Every entry of D is a $\mathbb{Z}[\zeta_p]$-combination of the d_v, each of valuation ≥ 1, so $D = \lambda D'$ with D' integral and $v_\lambda(\det D) \geq q$. As $v_\lambda(\det F) = 0$ we get $v_\lambda(e_q) \geq q$, and $q \geq v_\lambda(q)+4 = f(p-1)+4$ holds for every $f \geq 2$ (at $q=9$: $9 \geq 8$; at $q = 27$: $27 \geq 10$; at $q = 25$: $25 \geq 12$; the margin only grows). For $f = 1$ the inequality reads $q \geq q+3$ and fails, which is why the primes retain a residual. □

Proof. Newton's identities and Lemmas 10, 14 give $v_\lambda(e_k) \geq v_\lambda(q)+4$ for $3 \leq k \leq q-1$: the terms $e_{k-1}p_1$ and $e_{k-2}p_2$ are products of two quantities of valuation $\geq v_\lambda(q)+2$ by Theorem 12, and p_i for $i \geq 3$ has valuation $\geq v_\lambda(q)+4$. Theorem 16 handles $k = 1, 2$ jointly and the hypothesis handles $k = q$. □

So the only case still carrying a hypothesis is q prime, where it asks for $v_\lambda(\det D) \geq q+3$. Lemma 11 gives $v_\lambda(e_q) \geq v_\lambda(q)+2$ and Newton's identities give $v_\lambda(\det D) \geq q+1$, both short by two. The measured value is much larger:

$$\min_c v_\lambda(\det D) = 2q \qquad (6,\ 10,\ 14 \text{ at } q = 3, 5, 7),$$

with equality generically, so the Newton polygon of $D' = D/\lambda$ is the straight line of slope -1 and every eigenvalue of D has valuation exactly 2. Since $2q \geq q+3$ for every odd prime, the single statement

Conjecture 18. Every eigenvalue of D has λ-valuation at least 2; equivalently $v_\lambda(\det D) \geq 2q$.

closes the prime case and with it the whole law. In terms of the elementary symmetric functions $e_k = e_k(D)$ the conjecture says exactly $v_\lambda(e_k) \geq 2k$ for all k, and all but the last case is a theorem.

Moreover the last case reduces to a single power-sum bound. Newton's chain gives $v_\lambda(e_q) \geq v_\lambda(p_q) - v_\lambda(q) = v_\lambda(p_q) - (q-1)$, so $v_\lambda(p_q) \geq 2q+2$ would deliver $v_\lambda(e_q) \geq q+3$ — exactly what the sharp law needs. The counting-plus-parity argument only gives $v_\lambda(p_q) \geq 2q$, but that bound is *not tight*, and the excess has an exact form:

$$v_\lambda\big(\operatorname{tr} D^m\big) \ =\ v_\lambda(q) + m + [m \text{ odd}] \ +\ v_\lambda(m!). \tag{3}$$

Erratum 19. (3) is *false at* $q = 3$. At $q = 3$ there are $(q^2-1)/2 = 4$ inverse-closed pairs with $q = 3$ choices each, so the section space has exactly 81 elements and the minimum can be computed by exhaustion. Doing so gives

$$v_\lambda(\operatorname{tr} D^m) = 4, 8, 8, \mathbf{12}, 12, \mathbf{16}, \mathbf{16}, 20, 20, \mathbf{24}, 24 \qquad (m = 2, \ldots, 12),$$

whereas (3) predicts $4, 8, 8, 10, 12, 14, 14, 20, 20, 22, 24$: the true minimum exceeds the stated value by 2 at $m = 5, 7, 8, 11$. These eleven exhaustive values were the finite-window evidence for

$$v_\lambda(\operatorname{tr} D^m) = 2\big(m + [m \text{ odd}]\big) \qquad (q = 3).$$

Theorem 21 below proves the corresponding minimum over all 81 sections for every $m \geq 2$. Since $2v_3(m!) = m - s_3(m)$ with s_3 the base-3 digit sum, the two expressions differ by $s_3(m) + [m \text{ odd}] - 2$, which vanishes exactly when $s_3(m) + [m \text{ odd}] = 2$. For $m \geq 2$ this is equivalently $m = 3^j$ with $j \geq 1$, or $m = 3^i + 3^j$ with $0 \leq i \leq j$. The prime-power tower therefore lies inside, but does not exhaust, the locus on which the two agree, which is why its four confirmed rungs $8, 20, 56, 164$ could not detect this.

At $q = 3$ the scope of the failure is therefore exact: agreement holds if and only if $s_3(m) + [m \text{ odd}] = 2$ for $m \geq 2$. Beyond $q = 3$ the question remains open. At $q = 5$ the law is *not attained* at $m = 10$ or $m = 14$: the sampled minimum sits exactly 2 above it — the same gap as at $q = 3$ — over 250 sections, and over 600 in a separate probe, while $m = 6, 8, 12$ attain it. Sampling cannot refute, since the true minimum is at most what a sample exhibits and $q = 5$ has 5^{12} sections; so $q = 5$ remains undecided. But the reading that $q = 3$ might be special — its section space being the smallest possible — is now the less likely of the two, and we withdraw it.

The governing asymmetry is worth stating plainly: *confirming* (3) at one (q, m) requires only a single attaining section together with the never-below check, which is cheap; *refuting* it requires the entire section space, which is available only at $q = 3$. Every confirmation recorded in this paper is of the cheap kind.

For even m the true $q = 3$ value already has an elementary derivation. Across the complete 81-section space the pair $(v_\lambda(e_2), v_\lambda(e_3))$ takes exactly four non-degenerate values. The finite computation first showed that each determines the measured vector for $m \leq 10$; Theorem 21 and its corollary below remove that finite qualifier.

Proposition 20. *Let $q = 3$ and let c be one of the 32 sections with $\det D = 0$ and $v_\lambda(e_2) = 4$. Then $\operatorname{tr}(D^m) = 0$ for every odd m, and $v_\lambda(\operatorname{tr} D^m) = 2m$ for every even m. In particular the exhaustive even-m minimum is $2m$.*

Proof. By Pass 473 $e_1 = \operatorname{tr} D = 0$, and $e_3 = \det D = 0$ by hypothesis, so the characteristic polynomial is $x^3 + e_2 x = x(x^2 + e_2)$. Its roots are 0 and $\pm\mu$ with $\mu^2 = -e_2$, whence $v_\lambda(\mu) = v_\lambda(e_2)/2 = 2$. Then $\operatorname{tr}(D^m) = \mu^m + (-\mu)^m$, which vanishes for odd m and equals $2\mu^m$ for even m; as 2 is a unit at $p = 3$, $v_\lambda(\operatorname{tr} D^m) = m\, v_\lambda(\mu) = 2m$. $\square$

Theorem 21 (complete $q = 3$ trace-valuation law)**.** *For every integer $m \geq 2$, the minimum over the complete 81-section space is*

$$\min_c v_\lambda\big(\operatorname{tr} D_c^m\big) = 2\big(m + [m \text{ odd}]\big).$$

The exponent $m = 1$ is excluded because every section has trace zero.

Proof. The exhaustive characteristic-polynomial image consists of the six cubics

$$x^3 - 9ax - 27b, \qquad (a, b) \in \{(0,0), (1,0), (2,0), (3,0), (3,1), (4,3)\}.$$

Put $x = 3y$, and let S_m be the mth power sum of the roots of $y^3 - ay - b$. Newton's identities give

$$S_0 = 3, \qquad S_1 = 0, \qquad S_2 = 2a, \qquad S_m = aS_{m-2} + bS_{m-3}.$$

Consequently

$$\operatorname{tr} D_c^m = 3^m S_m, \qquad v_\lambda(\operatorname{tr} D_c^m) = 2m + 2v_3(S_m)$$

whenever the trace is nonzero.

For even m, integrality gives the lower bound $2m$. The row $(a,b)=(1,0)$ has $S_m = S_{m-2}$ and $S_2 = 2$, so $S_m = 2$ for every even $m \geq 2$ and the bound is attained.

For odd m, every $b=0$ row vanishes. For the two remaining rows write

$$\begin{aligned} A_m &= 3A_{m-2} + A_{m-3}, \quad (A_0, A_1, A_2) = (3,0,6), \\ B_m &= 4B_{m-2} + 3B_{m-3}, \quad (B_0, B_1, B_2) = (3,0,8). \end{aligned}$$

Modulo 3, every odd term of both recurrences is zero, so every finite odd trace has valuation at least $2(m+1)$. Modulo 9, the A-sequence is periodic from $m=0$ with word

$$3, 0, 6,$$

whereas the B-tail from $m=2$ has period six with word

$$8, 0, 5, 6, 2, 3.$$

Thus A_m is 3 or 6 modulo 9 for odd $m \equiv 3, 5 \pmod 6$, while $B_m \equiv 3 \pmod 9$ for $m \equiv 1 \pmod 6$, beginning at $m=7$. These cover every odd $m \geq 3$, and in each residue class one row has $v_3(S_m) = 1$. Hence the odd lower bound is also attained. The repeated recurrence states, rather than a finite sweep, prove the assertion for all m. □

The coefficient-valuation profile is consequently a complete invariant of the trace-valuation sequence for every m on the realized $q=3$ image. Its only repeated fibre is $(a,b) = (1,0), (2,0)$; in both rows the odd sums vanish and $S_{2k} = 2a^k$ is a 3-adic unit. The parity term is therefore not a correction applied to one minimizing section: the even minimum is supplied by the $b=0$ unit row, while the $(6,6)$ and $(4,8)$ profiles supply the odd residue classes. Combining the theorem with Legendre's identity makes the agreement locus with (3) an exact result:

$$(3) \text{ agrees at } m \geq 2 \quad \Longleftrightarrow \quad s_3(m) + [m \text{ odd}] = 2.$$

Equivalently, the agreement exponents are exactly

$$m = 3^j \ (j \geq 1) \qquad \text{or} \qquad m = 3^i + 3^j \ (0 \leq i \leq j).$$

Indeed $m \equiv s_3(m) \pmod 2$: the odd branch forces $s_3(m) = 1$, while the even branch forces $s_3(m) = 2$. A base-3 integer of digit sum 2 has either one digit 2 or two digits 1, both represented by $3^i + 3^j$ when equality of i and j is allowed. Nothing in this argument determines the minimum at $q \geq 5$. Statements below that invoke (3) remain conditional there, and the reduction of §4.5 is unaffected: the transfer-matrix identity and $T \equiv 0 \pmod \lambda$ are exact regardless.

The leading term is $v_\lambda(q)$, not $q-1$; those agree only when q is prime, and writing $q-1$ makes (3) fail by a constant 20 already at $|R| = 27$, where $v_\lambda(27) = 6$. The first three terms are the parity bound; the excess over it is precisely $v_\lambda(m!)$. This was found by separating two readings that prime q cannot distinguish. Over $\mathbb{F}_9$ and $\mathbb{F}_3[x]/(x^2)$ — both of block size 9 and characteristic 3, and giving *identical* profiles — the excess is not a single jump at $m = |R|$; it steps at $m = 3, 6, 9$, at multiples of the *characteristic*, with values $2, 2, 2, 4, 4, 4, 8$ for $m = 3, \ldots, 9$, matching $v_\lambda(m!)$ in every case. For prime q the only multiple of $p = q$ in range is $m = q$ itself, which is why a single jump at the end is all one sees there; and since $v_\lambda(q!) = q-1$ for prime q, (3) collapses to $v_\lambda(\operatorname{tr} D^q) = 3q-1$, reproducing $8, 14, 20, 32$ at $q = 3, 5, 7, 11$.

Since $3q - 1 \geq 2q + 2$ for every $q \geq 3$, this settles the residual numerically.

We do *not* claim to know why (3) holds. One tempting story — that the $m!$ is what Newton's identities divide by — is wrong: Newton computes the e_k *from* the p_m, whereas the $p_m = \operatorname{tr} D^m$ are primary traces computed with no division at all, so no factorial can enter that way. A mechanism with the right shape is the classical congruence $\operatorname{tr}(M^{p^k}) \equiv \operatorname{tr}(M^{p^{k-1}}) \pmod{p^k}$ for integral M, which with $\operatorname{tr} D = 0$ forces $\operatorname{tr}(D^{p^k}) \equiv 0 \pmod{p^k}$ and accumulates exactly as Legendre's $v_p(m!) = \sum_i \lfloor m/p^i \rfloor$; but we have verified only its first step, and do not assert it accounts for the whole profile. One ingredient *is* now proved — the [m odd] bracket, in §4.2 below.

Two further facts sharpen this. First, (3) *implies* the determinant law. At $m = q$ it reads $v_\lambda(\operatorname{tr} D^q) = 2q + v_\lambda(q!)$; Newton's chain then gives $v_\lambda(e_q) \geq q + 1 + v_\lambda(q!)$, and the law needs $v_\lambda(e_q) \geq q + 3$, so the entire residual is the inequality

$$v_\lambda(q!) \;\geq\; 2,$$

which for prime q is $q - 1 \geq 2$. Outside the now-settled $q = 3$ replacement, **the open problem is therefore no longer a statement about symplectic character sums at the top exponent; it is (3) itself**, an identity about the whole Newton recursion. Second, (3) has been tested at $|R| = 27$ over both $\mathbb{F}_{27}$ and $\mathbb{F}_3[x]/(x^3)$, matching *exactly* at all 26 exponents and across all nine increments of $v_\lambda(m!)$ — including the double steps at $m = 9, 18$ and the triple at $m = 27$ predicted by Legendre's formula.

4.2 One ingredient, proved: the odd-class vanishing

Expanding $\operatorname{tr}(D^m)$ over m-tuples $(v_1, \ldots, v_m)$ of total displacement zero, the summand is invariant under cyclic rotation, so the trace decomposes over the cyclic orbits,

$$\operatorname{tr}(D^m) \;=\; \sum_{d|m} d\, S_d, \qquad S_d = \sum_{\text{orbits of minimal period } d} \text{value at a representative.}$$

The following says that at odd m most of these classes are not merely small but identically zero.

Theorem 22 (odd-class vanishing)**.** *Write $m = dk$. If m is odd and $p \mid k$, then $S_d = 0$ identically, for every inverse-closed section.*

The hypothesis is not really about p. Both ingredients below use p only through the order e of the generating character, which over $\mathbb{F}_q$ happens to equal p: step (i) needs $kv = 0$ for every v, and step (ii) needs $\operatorname{ord}(u) \mid k$ for every u in the image of ψ — and each of those is $e \mid k$. The theorem therefore holds verbatim over any finite Frobenius ring with generating character of order e, with $p \mid k$ replaced by $e \mid k$; and over $\mathbb{Z}/9$, where $e = 9$, the constant class is measured to vanish at $m = 9$ and $m = 27$ and *not* at $m = 3$, exactly where "$p \mid m$" would have predicted vanishing. The order of the generating character is the same invariant that governs the scope of the determinant law itself (§5): two statements proved and tested separately are indexed by the same number.

Proof. Since $m = dk$ is odd, both d and k are odd. Let $(v_1, \ldots, v_d)$ represent an orbit of minimal period d and put $M = \rho_{v_1} \cdots \rho_{v_d}$. By the Heisenberg cocycle $M = \zeta^s \rho(w)$ with $w = \sum_i v_i$, so $M^k = \zeta^{ks}\rho(kw)$; as $p \mid k$ this kills both the phase and the argument, giving $M^k = I$ and $\operatorname{tr}(M^k) = q \in \mathbb{Z}$.

The coefficient of the orbit is $\prod_{i \le d}(u_i - 1)^k$ with each u_i a p-th root of unity. For $u = e^{2\pi \mathrm{i} j/p}$ one has $u - 1 = e^{\mathrm{i}\pi j/p}\, 2\mathrm{i}\sin(\pi j/p)$, hence

$$(u-1)^k = e^{\mathrm{i}k\pi j/p}\,(2\mathrm{i})^k \sin^k(\pi j/p),$$

which is purely imaginary exactly when $\frac{k-1}{2} + \frac{kj}{p} \in \mathbb{Z}$ for every j, i.e. exactly when k is odd and $p \mid k$ — our hypothesis. A product of d purely imaginary numbers is purely imaginary because d is odd.

Finally, inverse closure gives $c(-v) = -c(v)$, hence $d_{-v} = \overline{d_v}$, so the orbit of $(v_1, \ldots, v_m)$ pairs with that of $(-v_1, \ldots, -v_m)$ and the two together contribute $q \cdot 2\,\mathrm{Re}(\text{purely imaginary}) = 0$. Summing over pairs gives $S_d = 0$. □

For $d = 1$ the hypothesis is also necessary, and the witness is explicit rather than found by search.

Proposition 23 (the converse at $d = 1$)**.** *Let $p \mid m$, so that constant orbits exist. Then $S_1 = 0$ for every inverse-closed section if and only if m is odd.*

Proof. Sufficiency is Theorem 22. For necessity, fix one inverse-closed pair $\{v_0, -v_0\}$ and take the section with $\psi(c(v_0)) = u \neq 1$, $c(-v_0) = -c(v_0)$, and $c(v) = 0$ for every other v; this is inverse-closed by construction. Every d_v vanishes except $d_{v_0} = u - 1$ and $d_{-v_0} = \bar{u} - 1$, so

$$S_1 \;=\; q\big[(u-1)^m + \overline{(u-1)^m}\big] \;=\; 2q\,\mathrm{Re}(u-1)^m,$$

which is nonzero exactly when $(u-1)^m$ is not purely imaginary — by the criterion in the proof of Theorem 22, exactly when m is even. □

For $d > 1$ the converse is verified on 28 cells and remains unproved.

Two remarks. First, $p \nmid m/d$ precisely when d carries the whole p-part of m, so at odd m the decomposition *collapses*: the surviving classes are indexed by the divisors of $m/p^{v_p(m)}$, and when m is a power of p exactly one class survives, the free class $d = m$. Numerically the free class then carries the entire valuation — verified at $m = p$ and at $m = p^2 = 9$, where both short classes die in all twelve sampled sections. So the clean behaviour at $m = p$ is not a small-case accident; it is the prime-power case.

This makes $m = p^j$ the one place where (3) can be read off with no inter-class cancellation to confound it, and there the Legendre tower is exact:

$$\begin{aligned} p = 3: &\quad v_\lambda(\operatorname{tr} D^3) = 8, \quad v_\lambda(\operatorname{tr} D^9) = 20, \quad v_\lambda(\operatorname{tr} D^{27}) = 56; \\ p = 5: &\quad v_\lambda(\operatorname{tr} D^5) = 14, \quad v_\lambda(\operatorname{tr} D^{25}) = 54, \end{aligned}$$

each equal to $v_\lambda(q) + m + 1 + v_\lambda(m!)$ — minima over 250 and 60 sampled sections respectively — and including the double increment at $m = 25$ and the *triple* increment at $m = 27$, where $v_3(27!) = 9 + 3 + 1 = 13$.

A single surviving class also means $\operatorname{tr}(D^m) = m\, S_m$ exactly, so what the orbit *weight* contributes separates from what the orbit *sum* contributes, with no enumeration: $v_\lambda(S_m) =$

$v_\lambda(\operatorname{tr} D^m) - v_\lambda(m)$, giving $6, 16, 50$ at $p = 3$ and $10, 46$ at $p = 5$. The weight supplies $v_\lambda(m) = j(p-1)$, that is j of the $\sum_i \lfloor m/p^i \rfloor$ Legendre terms — all of them at $j = 1$, half at $j = 2$, and $3/13$ at $j = 3$. So the orbit mechanism accounts for the first Legendre increment and a vanishing share thereafter. That is a quantified limit on the mechanism, not an account of the remainder.

Second, the hypothesis of Theorem 22 is the bracket $[m \text{ odd}]$ of (3). That term — the one ingredient of the law that had the look of a fitted correction — is exactly the condition under which these classes vanish, and Theorem 22 is the first ingredient of (3) with a proof rather than a measurement. It uses inverse closure, the same hypothesis that rescues the first power sum (Lemma 14) and the top exterior power.

Theorem 22 says nothing when m is even. Both it and the even case are consequences of one statement.

Theorem 24 (the sieve)**.** *Let e be the order of the generating character and let $t \mid m$ satisfy: m/t is odd and $e \mid (m/t)$. Then*

$$\sum_{d \mid t} d\, S_d \;=\; q\Big(\sum_{v \neq 0} d_v^{m/t}\Big)^t \;=\; 0.$$

Proof. Put $n = m/t$. For $d \mid t$ we have $n \mid (m/d)$, hence $e \mid (m/d)$, so a period-d representative $(w_1, \ldots, w_d)$ has $M^{m/d} = \zeta^{s\,m/d} \rho\big((m/d)w\big) = I$ and the zero-sum condition $(m/d) \sum_i w_i = 0$ is automatic. So $\operatorname{tr}(M^{m/d}) = q$ and the orbit's value is $q \prod_{i \le d} d_{w_i}^{m/d}$, which is $\prod_{i \le t} d_{w_i}^{n}$ read over the repeated t-tuple. Since $d\, S_d$ counts each period-d orbit once per distinct t-tuple in it, summing over $d \mid t$ sweeps every t-tuple exactly once and gives $q\big(\sum_v d_v^n\big)^t$. The bracket vanishes because n is odd and $e \mid n$: $(u-1)^n$ is purely imaginary and inverse closure pairs v with $-v$. $\square$

Write $T = \{\, t \mid m : \; m/t \text{ odd}, \; e \mid (m/t) \,\}$. For m *odd*, T is exactly the set of divisors of m/e — downward closed — so Möbius inversion runs: $t = 1$ gives $S_1 = 0$, and then $t\, S_t = -\sum_{d \mid t,\, d < t} d\, S_d = 0$ for every $t \mid (m/e)$. That is precisely the hypothesis of Theorem 22, so the odd-class vanishing theorem is a corollary. For m *even*, T is not downward closed — at $m = jp$ only $t = j$ survives — and Theorem 24 degenerates to the single relation $\sum_{d \mid j} d\, S_d = 0$. One theorem covers both, and the difference between the odd and even cases becomes the question of whether a divisor set is downward closed.

At $m = 2p$ the relation reads $S_1 + 2S_2 = 0$ identically — verified exactly at $(p, m) = (3, 6)$, $(5, 10)$, $(7, 14)$ — so the residual there lives entirely in the class $d = p$, which $t = 2$ does not reach. This corrects a reading in an earlier draft, which lumped $d = 3$ together with $d = 1, 2$ at $(3, 6)$ and concluded that the short aggregate generically sits below the total; the aggregate over $d \mid t$ is not merely small but zero.

When $|T| > 1$ the theorem gives constraints neither of its two corollaries reaches. At $m = 18$ with $e = 3$ one has $T = \{2, 6\}$, and subtracting the two relations leaves

$$3S_3 + 6S_6 = 0,$$

a relation at *even* m that is not an individual vanishing: neither S_3 nor S_6 is zero on its own in any section sampled.

4.3 How much the sieve pins

Since Pass 510 the standing verdict has been that the orbit account is *partial*. Theorem 24 makes that a dimension count.

Proposition 25. *The sieve system on the unknowns* $\{S_d : d \mid m\}$ *has rank* $|T|$*. Hence, with* $U = \{u \mid m : \ u \text{ odd}, \ e \mid u\}$*,*

$$\mathrm{pinned}(m) = |U|, \qquad \mathrm{free}(m) = \tau(m) - |U|,$$

and writing $m = 2^b s$ *with* s *odd,* $\mathrm{free}(m) = (b+1)\tau(s) - \tau(s/e)$ *when* $e \mid s$*, and* $\tau(m)$ *otherwise.*

Proof. Relation t involves exactly the S_d with $d \mid t$, and S_t occurs in relation t and in no relation indexed by a proper divisor of t. So along any linear extension of divisibility the system is triangular with nonzero diagonal entries t, and its rank is the number of relations, $|T| = |U|$. The closed form is the divisor count $|U| = \tau(s/e)$ when $e \mid s$ and 0 otherwise, together with $\tau(m) = (b+1)\tau(s)$. □

Checked against the rank of the actual matrix over $\mathbb{Q}$ for $e \in \{3, 5, 7, 9, 25, 27\}$ and $m \leq 60$.

One may ask whether some further symmetry of the summand supplies relations the sieve misses. That is decidable rather than arguable: measure the classes themselves across many sections, expand each into its coordinates, and take the rank. The nullity is then the dimension of the space of universally valid linear relations with constant coefficients.

Erratum 26. An earlier draft reported that this nullity equals $|T|$ at $(p, m) = (3, 3), (3, 6), (3, 9), (3, 15), (5, 5), (7,$ and concluded that the sieve is not merely a family of relations but all of them. *The sieve is not complete.* Choosing cells where $|T| = 0$, so that the sieve predicts no relations at all, the measured nullity is 1 — at $(3, 2)$, $(3, 4)$, $(3, 10)$, $(5, 4)$ and $(5, 6)$ alike. The missing relation is elementary: the period-one class consists of the constant m-tuples $(v, \ldots, v)$, whose zero-sum condition is $mv = 0$ with $v \neq 0$, solvable exactly when $p \mid m$; so for $p \nmid m$ the class is *empty* and $S_1 = 0$ holds vacuously. The sieve counts relations forced by cancellation and does not count vacuous classes. The corrected statement is

$$\mathrm{nullity} \ = \ |T| + [\, p \nmid m \,],$$

verified on eleven cells across both regimes.

Every cell used in the earlier draft satisfies $p \mid m$, so the correction term vanished in all of them. The cells were not chosen adversarially: the shortcut of §4.4 requires $e \mid (m/d)$, a condition on p dividing m, so the affordable cells were exactly the cells where the claim cannot fail. Convenience and the blind spot were the same constraint — the second such instance in this paper, after the prime-power tower and (3).

The scope of what remains is worth stating plainly: relations whose coefficients depend on the section, and nonlinear relations, are not excluded. Adding sampled rows can only raise rank and lower nullity, so the sampled nullity is an upper bound. On the listed cells the explicit sieve/vacuous relations give the matching lower bound, making those equalities exact; a uniform all-cell law is not proved.

At a prime power $m = p^j$ the count is free $= 1$: the sieve pins j of the $j+1$ classes and leaves only the free class $d = m$. So the whole Legendre tower above the first increment sits in a single class, in which no cancellation between classes can occur; the orbit weight supplies j of the $\sum_i \lfloor m/p^i \rfloor$ Legendre terms and the orbit sum must supply the remaining $\sum_i \lfloor m/p^i \rfloor - j$ of them — that is $0, 2, 10, 36$ terms at $p = 3$ for $j = 1, 2, 3, 4$, or equivalently $(p-1)$ times that, $0, 4, 20, 72$ orders of λ. That is the sharpest form we have of what remains unproved. The tower itself is exact at every rung measured, as minima over sampled sections:

$$\begin{array}{llllll} p = 3: & 8, & 20, & 56, & 164 & (m = 3, 9, 27, 81); \\ p = 5: & 14, & 54 & & & (m = 5, 25); \\ p = 7: & 20, & 104 & & & (m = 7, 49). \end{array}$$

The sieve also survives where the determinant law does not. Its proof uses only $e \mid (m/d)$, which holds over $\mathbb{Z}/p^n$ exactly as over a field, and nothing in it uses (3); the relations are verified to hold at $\mathbb{Z}/9$, $\mathbb{Z}/25$ and $\mathbb{Z}/27$. That separates the structural facts that survive the failure from the arithmetic one that does not.

It is tempting to go further and read the failure depths $12, 30, 36$ as a shadow of how much the sieve leaves free. *They are not.* At the relevant exponent $m = q$ the free counts are $2, 2, 3$ at $\mathbb{Z}/9, \mathbb{Z}/25, \mathbb{Z}/27$, so $\mathbb{Z}/9$ and $\mathbb{Z}/25$ share a free count while their depths differ by 18; no function of the free count alone can produce both. We record the candidate as refused. The depths still match $p^{n-1}(p+1)$ exactly and remain unexplained.

4.4 A closed form, and the sieve in two lines

The classes need not be enumerated at all. Write $\mathrm{Ps}(k) = \sum_{v \neq 0} d_v^k$.

Proposition 27. *Whenever $e \mid (m/d)$,*

$$d\, S_d \;=\; q \sum_{c \mid d} \mu(d/c)\, \mathrm{Ps}(m/c)^{\,c}.$$

Proof. Put $x_v = d_v^{m/d}$. By the shortcut above, $d\, S_d$ is q times the sum of $\prod_i x_{w_i}$ over d-tuples of exact period d. A d-tuple has period dividing $c \mid d$ exactly when it is a c-tuple repeated d/c times, and then its product is $\left(\prod_{i \le c} x_{w_i}\right)^{d/c}$; summing over all c-tuples gives $\left(\sum_v x_v^{d/c}\right)^c = \mathrm{Ps}(m/c)^c$. Möbius inversion over the divisor lattice converts "period dividing" into "period exactly". $\square$

This gives Theorem 24 again, in two lines: summing over $d \mid t$ and exchanging the order of summation,

$$\sum_{d \mid t} d\, S_d = q \sum_{c \mid t} \mathrm{Ps}(m/c)^c \sum_{c \mid d \mid t} \mu(d/c) = q\, \mathrm{Ps}(m/t)^t,$$

the inner sum being 1 at $c = t$ and 0 otherwise. The counting proof given above and this one are both correct; this one says where the theorem comes from.

Computationally the difference is decisive. A class now costs $\tau(d)$ power sums, each a single loop over the $q^2 - 1$ vectors, in place of $|V|^d$ tuples — so $(7, 49)$ at $d = 7$, which would have needed 48^7 tuples, and $(3, 27)$ at $d = 9$, which would have needed 8^9, become immediate.

The hypothesis $e \mid (m/d)$ is not an artefact, and what it excludes can be named exactly. The representation satisfies

$$\rho(v)\, \rho(w) = \zeta^{-\omega(v,w)} \rho(v + w), \qquad \omega\big((a, b), (a', b')\big) = ab' - a'b,$$

so a period-d representative gives $M = \rho_{w_1} \cdots \rho_{w_d} = \zeta^{-\Omega}\rho(w)$ with $w = \sum_i w_i$ and $\Omega = \sum_{i<j} \omega(w_i, w_j)$; and since $\omega(w, w) = 0$ one has $\rho(w)^k = \rho(kw)$ exactly. With $k = m/d$ the zero-sum condition on the m-tuple is precisely $kw = 0$, so $\mathrm{tr}(M^k) = q\,\zeta^{-k\Omega}$ and the orbit's value is

$$q\,\zeta^{-k\Omega} \prod_{i \le d} d_{w_i}^k.$$

The shortcut above is this formula with its phase dropped, and the phase is trivial for *every* orbit exactly when $e \mid k$. That is why the closed form stops at the top class, where $k = 1$: not a gap in the argument but the shape of the object.

4.5 The transfer matrix

The phase does more than obstruct: it organises. Writing $P_j = v_1 + \cdots + v_j$ for the partial sums, the exponent telescopes,

$$\Omega = \sum_{i<j} \omega(v_i, v_j) = \sum_j \omega(P_{j-1}, v_j),$$

because $\sum_{i<j} \omega(v_i, v_j) = \sum_j \omega\big(\sum_{i<j} v_i,\, v_j\big)$. So the weight of an m-tuple factorises along the walk it traces through R^2, and the whole trace becomes a single matrix entry.

Theorem 28 (transfer matrix)**.** *Let T be the $q^2 \times q^2$ matrix on partial-sum states given by*

$$T[P + v,\ P] = d_v\, \zeta^{-\omega(P,v)} \qquad (v \neq 0).$$

Then $\mathrm{tr}(D^m) = q\,[T^m]_{0,0}$ for every m and every section.

Proof. $\rho(v_1) \cdots \rho(v_m) = \zeta^{-\Omega}\rho\big(\sum_i v_i\big)$ by the cocycle, and $\mathrm{tr}\,\rho(x) = q$ for $x = 0$ and 0 otherwise. So $\mathrm{tr}(D^m)$ is q times the sum, over m-tuples with $\sum_i v_i = 0$, of $\zeta^{-\Omega} \prod_i d_{v_i}$. By the telescoping identity that summand is $\prod_j d_{v_j} \zeta^{-\omega(P_{j-1}, v_j)}$, the product of the edge weights along the walk $0 = P_0, P_1, \ldots, P_m = 0$; summing over closed walks of length m based at 0 is $[T^m]_{0,0}$. □

Theorem 29 (translation covariance)**.** *For $a \in R^2$ let S_a permute the states by $P \mapsto P + a$ and let U_a be diagonal with entries $\zeta^{\omega(a,P)}$. Then $U_a T = S_a T S_a^{-1} U_a$ for every a. Consequently $[T^m]_{P,P} = [T^m]_{0,0}$ for every P, and $\mathrm{tr}(T^m) = q\ \mathrm{tr}(D^m)$.*

Proof. Both sides send the basis vector at P to a multiple of the one at $P + v$; the multipliers are $d_v \zeta^{\omega(a,P+v) - \omega(P,v)}$ and $d_v \zeta^{-\omega(P-a,v) + \omega(a,P)}$, equal since ω is bilinear and alternating. Conjugation by $S_a U_a$ therefore carries the $(0, 0)$ entry of T^m to the (a, a) entry, so the diagonal is constant; summing it gives $\mathrm{tr}(T^m) = q^2[T^m]_{0,0} = q\ \mathrm{tr}(D^m)$ by Theorem 28. □

This explains the factor q in Theorem 28: it is not a normalisation accident but the count of states in a translation orbit.

It is tempting to go further and say that $E(m) = v_\lambda([T^m]_{0,0}) - m$ is therefore a similarity invariant of T, so that computing T's spectrum would settle every m at once. *That is false, and Theorem 29 is what refutes it.* Power sums determine a multiset of eigenvalues, and $\mathrm{tr}(T^m) = q\ \mathrm{tr}(D^m)$ for every m forces T's spectrum to be D's with each eigenvalue repeated q times — exactly $q \cdot q = q^2$ of them, filling the matrix with no zeros. Equivalently

$$\mathrm{charpoly}(T) = \mathrm{charpoly}(D)^{\,q},$$

verified exactly at $q=3$. So T carries precisely the spectral information D carries and no more; computing it computes what was already available from D, and what §4.5 buys is a language for the orbit results — cycle types, necklace counts, the sieve as Möbius inversion over walks — not a new invariant. A reformulation that preserves all the information preserves all the difficulty.
Everything in this section is a statement about closed walks in this weighted graph, grouped by rotation of the base point: the period-d classes are cycle types, Theorem 24 is a Möbius inversion over them, and Proposition 27 is the necklace count.

The reformulation removes one summand of (3) outright. Every entry of T is a $\mathbb{Z}[\zeta]$-combination of the d_v, and $v_\lambda(d_v) \geq 1$ for every v; hence $T \equiv 0 \pmod{\lambda}$ and

$$v_\lambda\big([T^m]_{0,0}\big) \;\geq\; m$$

with no work at all. Since $\mathrm{tr}(D^m) = q[T^m]_{0,0}$ contributes the $v_\lambda(q)$, what remains of (3) is exactly the excess

$$E(m) := v_\lambda\big([T^m]_{0,0}\big) - m = [m \text{ odd}] + v_\lambda(m!),$$

a statement about one entry of one matrix. Theorem 21 determines the minimum of this entrywise excess at $q=3$, where the factorial expression is false. For the unresolved larger-q cases we do not prove the displayed identity; this is the remaining spectral-arithmetic step.

A corollary of the sieve is worth recording in this language. Taking $t = m/e$ gives $\sum_{d|m/e} d\, S_d = 0$, so

$$\mathrm{tr}(D^m) = \sum_{\substack{d|m \\ d\nmid m/e}} d\, S_d :$$

the trace is carried entirely by the classes whose period does *not* divide m/e. At $m = p^j$ the only such class is $d = m$, recovering $\mathrm{tr}(D^m) = m\, S_m$; at $(p,m) = (3,6)$ the carriers are $d=3$ and $d=6$.

Proposition 27, together with the vacuous-class correction of Erratum 26, settles which m have a single free class.

Proposition 30. $\mathrm{free}(m) = 1$ *if and only if* m *is a positive power of* p *or a prime different from* p*; moreover* $\mathrm{free}(1) = 0$.

Proof. Use the corrected count $\mathrm{free}(m) = \tau(m) - |T| - [p \nmid m]$. If $p \mid m$, the last term vanishes and the divisor count gives $\mathrm{free}(m) = 1$ exactly for $m = p^a$, $a \geq 1$. If $p \nmid m$, then T is empty and $\mathrm{free}(m) = \tau(m) - 1$; this is 1 exactly when m is prime, and is 0 at $m = 1$. □

Remark 31 (Historical correction)**.** An earlier version of Proposition 30 asserted only the positive powers of p. Its input $\mathrm{free}(m) = \tau(m) - |U|$ omitted the vacuous period-one class of Erratum 26. The extra term $-[p \nmid m]$ supplies the second family, the primes $\ell \neq p$.

A proof inherits the blind spot of the formula it starts from. This is the third instance in this paper; the failure mode itself, with all three examples, is written up separately in `papers/agreement_locus.tex`.

We also record the status of Erratum 26's corrected law more precisely than "measured". Its lower bound nullity $\geq |T| + [\,p \nmid m\,]$ is proved: both relation families are genuine and, since $|T| \geq 1$ exactly when $p \mid m$, they never coexist. Its upper bound comes from exhibiting sections whose class vectors are linearly independent over $\mathbb{Q}(\zeta_p)$ — and exhibiting independent vectors *proves* a rank lower bound rather than sampling one, the independence being exact arithmetic. So for each cell where the two bounds meet, the equality is a theorem about that cell; what remains open is the statement for all m simultaneously, which no finite computation can settle.

The closed form also lets the completeness test be redone with coefficients in $\mathbb{Q}(\zeta_p)$ rather than $\mathbb{Q}$, by carrying out the elimination in the field itself — inverting through the multiplication matrix on the reduced power basis. The nullity again equals $|T|$ — every one of these cells has $p \mid m$, so by Erratum 26 the correction term $[\, p \nmid m \,]$ vanishes throughout and the agreement is expected — now at $(p, m) = (3, 6), (3, 9), (3, 15), (3, 27), (3, 81), (5, 25), (7, 49)$, the last four of which only the closed form makes computable. One caution the first run supplied: at odd m a large fraction of sections give $\mathrm{tr}(D^m) = 0$ identically, so their entire class vector vanishes and contributes nothing to the rank. Sampling a fixed range of sections drew only such sections at $m = 15, 27, 81$ and returned rank 0, which looks like a missing relation and is really an empty sample; the figures above count *informative* sections.

Finally, the proof's first step is a useful computational tool in its own right: whenever $e \mid (m/d)$, a period-d orbit's value is $q \prod_i d_{w_i}^{m/d}$ with no matrix products at all. Checked against the honest matrix computation on 15,616 orbits over $\mathbb{F}_3, \mathbb{F}_5$ and on the constant class over $\mathbb{Z}/9$, it is what puts $\mathbb{Z}/25$ and $\mathbb{Z}/27$ — 624 and 728 vectors carrying 25×25 and 27×27 blocks — within reach, and there the character-order form of the hypothesis is confirmed at a second p^2 and a first p^3.

The orbit account is therefore partial for a sharper reason than "the even case is untouched": what remains unexplained at $m = jp$ is exactly the classes with $d \nmid j$. At $(3, 6)$ the total ranges over $12, 14, 18$ by section, with minimum $12 = v_\lambda(3) + 6 + 0 + v_\lambda(6!)$, the value predicted by (3) and consistent with the law being a minimum over sections.

Finally, (3) fails — and fails *from below* — exactly where the determinant law fails. Over $\mathbb{Z}/9$ and $\mathbb{Z}/25$, whose generating characters have order p^2, the measured power sums lie strictly under the right-hand side (at $\mathbb{Z}/9$, $m = 9$: 30 against 46). So the two failures share a locus and presumably a cause, and (3) is not a more robust statement than the law it implies.

Theorem 22 sharpens that observation from a shared locus to a named broken hypothesis. Ingredient (i) needs $\rho_v^p = I$, which came from $(v, 0)^p = (pv, 0)$ being the identity in characteristic p; over $\mathbb{Z}/p^n$ it is not, and the failure is severe — over $\mathbb{Z}/9$ only 8 of the 80 nonzero vectors satisfy $\rho_v^3 = I$ (exactly those with $3v = 0$), and over $\mathbb{Z}/25$ only 24 of 624. Correspondingly the constant class at $m = p$, which Theorem 22 kills over $\mathbb{F}_q$, is measured to be nonzero over both rings. So in the failure region a specific hypothesis of the proof is false and the corresponding conclusion is false. We record this as a *candidate*: it identifies where the argument breaks, and it does not establish that this is why (3) fails.

Theorem 32 (the eigenvalue bound, off the top)**.** *For every odd prime q and every inverse-closed section, $v_\lambda(e_k) \geq 2k$ for all $1 \leq k \leq q - 1$.*

Proof. Induction on k. Since $p_1 = \mathrm{tr}\, D = 0$, Newton's identity reads $k\, e_k = \sum_{i=2}^{k} (-1)^{i-1} e_{k-i} p_i$. By Lemma 10 $v_\lambda(p_i) \geq v_\lambda(q) + i$, and by induction $v_\lambda(e_{k-i}) \geq 2(k - i)$, so each term has valuation at least

$$2(k - i) + v_\lambda(q) + i \;=\; 2k - i + v_\lambda(q) \;\geq\; 2k,$$

because $i \leq k \leq q - 1 = v_\lambda(q)$. For $k \leq q - 1$ the integer k is a unit at λ, so dividing by it is harmless. □

The induction fails at $k = q$ for one identifiable reason: there i may equal q, and the inequality $i \le v_\lambda(q) = q-1$ misses by exactly one. Accordingly Newton yields only $v_\lambda(\det D) \ge q+1$ against the measured $2q$. So the entire residual gap in the determinant law is the single coefficient $e_q = \det D$ of one characteristic polynomial.

Two further pieces of evidence. First, $e_2 = qS$ exactly, from $e_2 = (p_1^2 - p_2)/2 = -p_2/2$ and $p_2 = -2qS$ (Lemma 15); this is the second exact closed form in the family after $\operatorname{tr} H = qS$. Second, D' reduces mod λ to $\sum_a \kappa_a P_a$ with $\kappa_{-a} = -\kappa_a$ (so $\kappa_0 = 0$ and $\sum_a \kappa_a = 0$), which lies in the augmentation ideal of $\mathbb{F}_q[\mathbb{Z}/q] = \mathbb{F}_q[s]/(s^q)$ — a nilpotent ideal. Hence every eigenvalue of D' is a nonunit, giving valuation > 0; the conjecture upgrades this to ≥ 1.

Remark 33. The exponent is attained: at $q = 3$ every non-flat section gives valuation exactly $6 = v_\lambda(q) + 4$, at $q = 5$ the observed values are $\{8, 10, 12\}$ with minimum 8, at $q = 7$ the minimum is 10, and at $q = 9$ — where $v_\lambda(9) = 4$, not $q - 1 = 8$ — the minimum is 8. All four agree with $v_\lambda(q) + 4$.

Remark 34. The residue class is the flat determinant, not a numerical coincidence of small q. At $q = 3$ the two determinant values are -16 (flat orbit) and 11, which differ by 27; since $(\lambda^6) = (27)$ for $q = 3$, the congruence $\det B \equiv \det F \pmod{\lambda^{q+3}}$ there reads $\det B \equiv -16 \pmod{27}$, and $11 \equiv -16$. An earlier guess that the residue is $q^2 + 2$ fails already at $q = 5$.

5 Prime powers

Let $q = p^f$ with $f > 1$, $\psi_t(z) = \zeta_p^{\operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(tz)}$, and $\lambda = 1 - \zeta_p$. Lemma 3 survives verbatim: at $q = 9$ one finds $F^2 + 2F - 80I = 0$, spectrum $\{8^5, (-10)^4\}$, and $\det F = 8^5(-10)^4 = 327{,}680{,}000$, in agreement with the formula of Lemma 3; the trace laws hold as well.

Nothing else changes. Every ingredient of the argument — the flat-block Lemma 3, the vanishing of $\operatorname{tr} \rho_t$ off the centre (Lemma 2), the pairing Lemma 9, the parity Lemma 14, and the two closed forms of Lemma 15 — holds verbatim with ψ_t in place of $\zeta^{t\,\cdot}$ and ω the symplectic form over $\mathbb{F}_q$. In particular the symplectic cancellations go through because $\sum_{x \in \mathbb{F}_q^2} \omega(x, v) = 0$ for $v \ne 0$ and because $q^2 \equiv 0$ in the residue field, neither of which sees f.

The single quantity that depends on f is the ramification,

$$v_\lambda(q) = v_\lambda(p^f) = f(p-1),$$

since $(p) = (\lambda)^{p-1}$ in $\mathbb{Z}[\zeta_p]$. Theorems 12, 16 and 17 were therefore already stated in their correct generality: *there is no separate prime-power theory.*

Remark 35 (why this is one law and not two)**.** At prime q we have $v_\lambda(q) = q - 1$, so the sharp exponent $v_\lambda(q) + 4$ reads $q + 3$; at $q = 9$ we have $v_\lambda(9) = 4$ and it reads 8. As functions of q these look unrelated — 8 sits far below the value $q + 3 = 12$ that the prime formula would predict — which is why we first recorded the two cases as different phenomena, the prime case as a law and the prime-power case as its failure. As functions of $v_\lambda(q)$ they are one statement. The constant $4 = 2 + 2$ counts inverse closure twice, once at first order and once in the e_1/e_2 cancellation, and is blind to the factorization of q; only the ramification varies.

The measured minima agree with $v_\lambda(q) + 4$ at every prime power tested, across three different values of f:

q (p,f)	3 $(3,1)$	5 $(5,1)$	7 $(7,1)$	9 $(3,2)$	25 $(5,2)$	49 $(7,2)$	27 $(3,3)$	81 $(3,4)$
$v_\lambda(q)$	2	4	6	4	8	12	6	8
$v_\lambda(q)+4$	6	8	10	8	12	16	10	12
observed min	6	8	10	8	12	16	10	12

Eight prime powers across four values of f; the last five columns are unconditional by Theorem 17. Note that q and $v_\lambda(q)$ order the columns differently — $q = 49$ has a larger exponent than $q = 81$ — which is the clearest sign that the ramification, not q, is the governing parameter. The determinants beyond $q = 9$ were computed by fraction-free (Bareiss) elimination over $\mathbb{Z}[\zeta_p]$, validated against cofactor expansion at $q = 3, 5, 7$; the flat determinants match Lemma 3 exactly in every case, including the 35- and 39-digit values $24^{13}26^{12}$ at $q = 25$ and $-26^{14}28^{13}$ at $q = 27$.

5.1 The law does not survive over $\mathbb{Z}/p^n$

It is natural to ask whether the field can be replaced by a finite chain ring. It cannot, and the failure is instructive. Take $q = 9$ but with the ring $\mathbb{Z}/9$ in place of $\mathbb{F}_9$: the group, the sections and the inverse-closure condition are unchanged, and only the central character changes, from order 3 to order 9. The construction remains valid — ρ is a homomorphism, $\operatorname{tr}\rho$ vanishes off the centre, and the flat block still satisfies $F^2 + 2F - 80I = 0$ with spectrum $\{8^5, (-10)^4\}$ and $\det F = 327{,}680{,}000$, numerically identical to the field case. But now $\lambda = 1 - \zeta_9$ and $v_\lambda(9) = 12$, so the law would predict exponent 16.

The measured exponent is $12 = v_\lambda(q)$: over 20 sections the depths are $\{12, 16, 18\}$. *The entire* $+4$ *is lost.* The mechanism is exact, and it is a single one. Newton's identity divides by k, which is harmless only when k is a λ-unit; and while $v_\lambda(3) = 2$ in $\mathbb{Z}[\zeta_3]$, in $\mathbb{Z}[\zeta_9]$ one has $v_\lambda(3) = 6$. With block size 9 the recursion passes through $k = 3, 6, 9$ in both cases, and only when the character has order 9 is the loss large enough to consume the whole $+4$.

It is tempting to blame the ring structure — $\mathbb{Z}/9$ has zero divisors and nonzero vectors that are not unimodular, so one might expect the symplectic sums of Lemma 15 to degrade. *They do not.* For any finite Frobenius ring R with generating character ψ one has $\sum_{a\in R}\psi(au) = 0$ for every $u \neq 0$ — that is the defining property — so

$$\sum_{x\in R^2} \psi\big(\omega(x,u)\big) = \Big(\sum_{x_0}\psi(x_0u_1)\Big)\Big(\sum_{x_1}\psi(-u_0x_1)\Big) = 0$$

for every $u \neq 0$, unimodular or not; we verified this exhaustively over $\mathbb{Z}/9$. The character sums are perfectly healthy over the ring.

Proposition 36 (the discriminator is the character order)**.** *What controls the* $+4$ *is the order of the generating character, not whether the coefficient ring is a field.*

The evidence is a third experiment, chosen to separate the two readings. Let $R = \mathbb{F}_3[x]/(x^2)$, a Frobenius ring which is *not* a field — it has zero divisors and non-unimodular nonzero vectors — but whose generating character $\psi(c) = \zeta_3^{c_1}$ (the socle coordinate) has order 3. Then $|R| = 9$ and $v_\lambda(9) = 4$, exactly as over $\mathbb{F}_9$. Running the same computation: ρ is a homomorphism, the flat block again satisfies $F^2 + 2F - 80I = 0$ with $\det F = 327{,}680{,}000$, and the measured depths are $\{8, 10, 12\}$ with minimum $8 = v_\lambda(q) + 4$. **The $+4$ survives over a non-field.**

So the trichotomy is: $\mathbb{F}_9$ (field, character order 3) gives 8; $\mathbb{F}_3[x]/(x^2)$ (not a field, character order 3) gives 8; $\mathbb{Z}/9$ (not a field, character order 9) gives 12. The field hypothesis plays no role; Theorem 17 is a statement about coefficient rings whose generating character has order p, and its proof goes through verbatim in that generality.

The family $R = \mathbb{F}_p[x]/(x^k)$ — socle (x^{k-1}), generating character $\psi(c) = \zeta_p^{c_{k-1}}$ of order p, $|R| = p^k$ — realizes this for every p and k, with $k = 1$ the field $\mathbb{F}_p$ and $k \geq 2$ a genuine non-field. We tested it along both axes:

R	$\lvert R\rvert$	field?	$v_\lambda(\lvert R\rvert)$	predicted	observed min
$\mathbb{F}_3[x]/(x^2)$	9	no	4	8	8
$\mathbb{F}_3[x]/(x^3)$	27	no	6	10	10
$\mathbb{F}_5[x]/(x^2)$	25	no	8	12	12

Each is checked with a homomorphism-validated representation, the flat-block quadratic, and the closed-form flat determinant. Commutativity is not needed to define the Heisenberg cocycle: it is associative over any ring (the six cross terms cancel identically) and the centre is central. Beyond the theorem scope used above, the law is confirmed exactly over two noncommutative Frobenius rings of order 81 — the semisimple $M_2(\mathbb{F}_3)$ with $\psi(X) = \zeta_3^{\operatorname{tr} X}$, and the *local* twisted dual numbers $\mathbb{F}_9[\theta]$, $\theta^2 = 0$, $\theta a = a^3\theta$, with ψ the socle character. Both have $v_\lambda(81) = 8$, in both the representation is a homomorphism and the flat determinant matches Lemma 3, and both attain minimum depth $12 = v_\lambda(|R|) + 4$. The second is the more informative: being local and nilpotent as well as noncommutative, it tests those features together, which the semisimple $M_2(\mathbb{F}_3)$ cannot. The exponents coincide exactly with those of the *fields* of the same size ($\mathbb{F}_{27} \to 10$, $\mathbb{F}_{25} \to 12$): **the law cannot distinguish $\mathbb{F}_{p^k}$ from $\mathbb{F}_p[x]/(x^k)$.** The measured bound $v_\lambda(\det D) \geq 2|R|$ of Conjecture 18 also holds over each of them, with equality (18, 54, 50), so the sole open step behaves identically off the field locus.

Finally, the hypothesis is *necessary*, not merely convenient. Repeating the $\mathbb{Z}/9$ experiment at a second prime, over $\mathbb{Z}/25$ — generating character of order 25, $v_\lambda(5) = 20$, $v_\lambda(25) = 40$, so Newton's divisions by $5, 10, 15, 20, 25$ each cost 20 against the 4 they cost over $\mathbb{Z}[\zeta_5]$ — the law would predict exponent 44 and the measured minimum is 30. So the law fails at both $p = 3$ and $p = 5$ once the character order exceeds p.

The failed value is *not* a function of $v_\lambda(q)$: $\mathbb{Z}/9$ landed exactly on $v_\lambda(q) = 12$, whereas $\mathbb{Z}/25$ gives 30, strictly below $v_\lambda(25) = 40$. A third point resolves the picture. At $\mathbb{Z}/27$ the minimum is 36, and

$$12,\ 30,\ 36 \;=\; q + \frac{q}{p} \;=\; p^{\,n-1}(p+1) \qquad \text{at } \mathbb{Z}/p^n = \mathbb{Z}/9,\ \mathbb{Z}/25,\ \mathbb{Z}/27.$$

So the failure region carries a law of its own — one agreeing with $v_\lambda(q) = n\,p^{n-1}(p-1)$ (namely $12, 40, 54$) only at $\mathbb{Z}/9$, which is precisely why two data points looked structureless.

Conjecture 37. Over $\mathbb{Z}/p^n$ with p odd, $\min_c v_\lambda\big(\det B_t(c) - \det F\big) = p^{\,n-1}(p+1)$.

Three data points; a proof would presumably come from tracking which Newton division dominates the recursion.

6 What is left

The machinery of the last sections — a cyclic decomposition, a sieve, a closed form, a transfer matrix and its covariance — all describes one object. Theorem 29 gives $\mathrm{charpoly}(T) = \mathrm{charpoly}(D)^q$, so the transfer matrix carries exactly the spectral information D carries; and the trace-valuation vector is a function of $\mathrm{charpoly}(D)$, verified on 160 distinct characteristic polynomials at $q = 5$ with none mapping to two vectors. Beyond $q = 3$ the remaining spectral problem is a single sentence:

$$\textit{given } \mathrm{charpoly}(D), \textit{ compute } v_\lambda\big(\textstyle\sum_j \nu_j^m\big).$$

At $q = 3$ the section-space image is answered. The 81 sections map onto exactly six characteristic polynomials, all of the form

$$x^3 - 9a\,x - 27b, \qquad (a, b) \in \{(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (4, 3)\},$$

with multiplicities $1, 8, 24, 8, 24, 16$; the image is six lattice points. They give six exact recurrences, and Theorem 21 computes their minimum for every $m \geq 2$. After the normalization $x = 3y$, the recurrence is $S_m = aS_{m-2} + bS_{m-3}$; the unit $b = 0$ row attains every even minimum, while the modulo-9 words $(3, 0, 6)$ and $(8, 0, 5, 6, 2, 3)$ cover the three odd residue classes modulo 6. The valuation profile used in §4.4 is the coarsening that merges $x^3 - 9x$ with $x^3 - 18x$. Those two rows nevertheless have the same complete valuation sequence, so the profile is a complete invariant for all m on the realized $q = 3$ image. The lattice containment is now proved: inverse closure makes D Hermitian, so its characteristic polynomial is real and hence rational at $p = 3$; then $v_\lambda(e_2) \geq 4$ and $v_\lambda(e_3) \geq 6$ are exactly $9 \mid e_2$ and $27 \mid e_3$. What remains unexplained is why precisely these six allowed lattice points occur.

The orbit structure sharpens the multiplicities. The 81 sections form seven $\mathrm{Sp}(2, 3)$-orbits of sizes $1, 8, 8, 8, 8, 24, 24$. Exactly two orbits merge spectrally: they are the two full-support orbits and share $x^3 - 36x - 81$. In the certificate's fixed oriented antipodal frame, their coordinate products are 1 and 2 in $\mathbb{F}_3$; intrinsically the separator is the product corrected by the frame factor in the Moore–Dickson coefficient. Reorienting one representative may swap the bare labels but not the unordered split. After identifying nonzero coordinates with signs, these are the two eight-vertex demicubes, equivalently the two chiral D_4 half-spin weight sets. A determinant--1 element of $\mathrm{GL}(2, 3)$ swaps them together with the two central characters; their Galois-conjugate Hermitian polynomials coincide because $\mathbb{Q}(\zeta_3)^+ = \mathbb{Q}$. Thus the merge is explained rather than merely located.

At $q = 5$, coefficient valuations still do not determine the trace vector, but finite sampling cannot determine the global image size. Burnside gives 2,034,735 total symplectic section orbits and, exactly, 139,904 full-support orbits. A deterministic sample of 3,000 distinct full-support orbits found 34 characteristic-polynomial merges: 33 are explained by determinant-twisted linear equivalence, while one pair is affine-inequivalent. Once found, that pair is checked exactly as a further full-support pair of cospectral, nonisomorphic Cayley graphs. It is outside the eight explicit affine pairs retained by Passes 456, 479, and 482 and realizes the sheet-coincidence mechanism of Pass 481; Pass 482 did not retain every sampled representative, so no global ordinal is claimed. This is an existence certificate, not an estimate of the number of characteristic polynomials or of collision density. The two reduced Laplacians also have identical critical groups: their nonunit Smith invariant factors are 5 with multiplicity 16, 25 with multiplicity 5, 125 with multiplicity 13, and 2028949923625 with multiplicity 10. Their unequal local profiles therefore separate graphs that both spectrum and critical group fail to distinguish.

The signed-cycle Burnside argument uses only that 2 is a unit, so it holds over *every* odd $\mathbb{Z}/p^n$ as one formula. There $\mathrm{SL}(2, \mathbb{Z}/p^n)$, of order $p^{3n-2}(p^2-1)$, permutes the $(p^{2n}-1)/2$ antipodal pairs with a sign; a negative cycle forces the section value to zero and a positive cycle is free, so $|\mathrm{Fix}_{\mathrm{all}}(g)| = (p^n)^{c_+(g)}$ and $|\mathrm{Fix}_{\mathrm{full}}(g)| = (p^n-1)^{c_+(g)}$ when g has no negative cycle and 0 otherwise, with $c_+(g)$ the number of positive cycles. The field counts of Lemma-scale interest—7 and 2,034,735 at $\mathbb{F}_3, \mathbb{F}_5$ with full-support 2 and 139,904—are the case $n=1$, and $\mathbb{Z}/9$ is $(p,n) = (3,2)$, giving 228100045392509153077600971330057241 section orbits and 2051277771273019233341050472890368 full-support orbits; the next rings $\mathbb{Z}/25$ and $\mathbb{Z}/27$ follow at once. These counts and the $q=3$ chirality statement are exhaustive; only the route by which the $q=5$ witness was located is sampled. The orbit count bounds the number of characteristic polynomials but does not equal it, as the $q=5$ full-support cospectral pair shows.

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